

1.1

Base 10 (decimal)	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64
Base 8 (octal)	62	63	64	65	66	67	70	71	72	73	74	75	76	77	100
Base 16 (hexadecimal)	32	33	34	35	36	37	38	39	3A	3B	3C	3D	3E	3F	40

Numbers from 10_{10} to 30_{10} in base 14:

Base 10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Base 14	B	C	D	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	20	21	22

1.2

- (a) 16384 bytes
- (b) 33554432 bytes
- (c) 2147483648 bytes

1.3

- (a) 99
- (b) 1179
- (c) 26490
- (d) 206

1.4

- Decimal: 4095
- Octal: 7777
- Hex: FFF

1.5

- (a) 13 base 5
- (b) 7 base 5
- (c) 4 base 3

1.6

- $x = 6$

1.7

- Octal: 145136

1.8

- (a) 1111_1101_{10}
- (b) $FD_{16} = 1111_1101_{10}$

1.9

- (a) 21.625
- (b) 100.5
- (c) 177.5625
- (d) 20958.75
- (e) 51.125

1.10

- (a) 1.09375
- (b) 8.75

1.11

- $101010 \div 100 = 1010.1$

1.12

- (a) 1101 and 110.
- (b) $D0_h$ and $1F_h$

1.13

- (a)

Fractional Portion of $45.125 = 0.125$

Multiplication Integer		Fraction	Coefficient
0.125×2	0	.25	$a_{-1} = 0$
0.25×2	0	.5	$a_{-2} = 0$
0.5×2	1	0	$a_{-3} = 0$

$$45.125_{10} = 101101.001_2$$

- (b)

$$3.33_{10} = 11.01010000_2$$

$$\text{Decimal of } 11.01010000_2 = 2 \times 1 + 1 + 0.25 + 0.0625 = 3.3125$$

$$\text{Error} = \frac{3.33 - 3.3125}{3.33} \times 100 = 0.526\%$$

- (c)

$$\text{Hexadecimal of } 11.01010000_2 = 3.50_{16} = 3 + \frac{5}{16} = 3.3125_{10}, \text{ same as (b).}$$

- 1.14 (a) Given number: 1110 0010
 1s complement: 0001 1101
 2s complement: 0001 1110
- (b) Given number: 0001 1000
 1s complement: 1110 0111
 2s complement: 1110 1000
- (c) Given number: 1011 1101
 1s complement: 0100 0010
 2s complement: 0100 0011
- (d) Given number: 1010 0101
 1s complement: 0101 1010
 2s complement: 0101 1011
- (e) Given number: 1100 0011
 1s complement: 0011 1100
 2s complement: 0011 1101
- (f) Given number: 0101 1000
 1s complement: 1010 0111
 2s complement: 1010 1000

- 1.15 (a) Given number: 65,234,035
 9s complement: 34,765,964
 10s complement: 34,765,965
- (b) Given number: 56,783,223
 9s complement: 43,216,776
 10s complement: 43,216,777
- (c) Given number: 87,000,367
 9s complement: 12,999,632
 10s complement: 12,999,633
- (d) Given number: 99,999,000
 9s complement: 00,000,999
 10s complement: 00,001,000

1.16 (a) 8's complement of 2360_8 :

7's complement of $2360_8 = 5417_8$

8's complement = 5420_8

(b) Binary equivalent of $2360_8 = 010_011_110_000_2$

(c) 2's complement of $010_011_110_000_2 = 101_100_010_000_2$

(d) Octal of $101_100_010_000_2 = 5420_8$ which is same as result obtained in (a).

1.17

- (a) 2956
- (b) -974
- (c) -488
- (d) 1445

1.18

- (a) 111
- (b) -110000
- (c) -110
- (d) 1010010

1.19

- (a) 010087
- (b) 008485
- (c) 991515
- (d) 989913

1.20

- (a) -20
- (b) +20
- (c) -78

1.21

- (a) 010383

- (b) 009101
- (c) -9101
- (d) -10383

1.22 Decimal = 9,045
 BCD = 1001_0000_0100_0101
 ASCII = 0_011_1001_0_011_0000_1_011_0100_0_011_0101

 Decimal = 337
 BCD = 0011_0011_0111
 ASCII = 0_011_0011_0_011_0011_0_011_0110

1.23

- 1125

1.24	(a)	5	2	1	1	Decimal
		0	0	0	0	0
		0	0	0	1	1
		0	0	1	1	2
		0	1	0	1	3
		0	1	1	1	4
		1	0	0	0	5
		1	0	0	1	6
		1	0	1	1	7
		1	1	0	1	8
		1	1	1	1	9

(b)	6	3	2	1	Decimal
	0	0	0	0	0
	0	0	0	1	1
	0	0	1	0	2
	0	0	1	1	3
	0	1	0	0	4
	0	1	0	1	5
	0	1	1	1	6
	1	0	0	1	7
	1	0	1	0	8
	1	0	1	1	9

1.25	(a)	6,428 ₁₀ BCD:	0110_0100_0010_1000
	(b)	Excess-3:	1001_0111_0101_1011
	(c)	2421:	1100_0100_0010_1110
		2421:	0110_0100_1000_1110
	(d)	6311:	1000_0110_0010_1011

1.26

6,428 9s Comp:	3,571
	6 4 2 8
2421 code:	0011_1011_0111_0001
1.25(c):	1100_0100_0010_1110 (2421 code – alternative #1)
1s comp:	0011_1011_1101_0001 (2421 code - alternative #2)
	6 4 2 8
6,428 ₂₄₂₁	0110_0100_1000_1110 (2421 code alternative #2)
1s comp	1001_1011_0111_0001 Match

5,736 9s Comp:	4,263
2421 code:	0100_0010_1100_0011
1s comp:	1011_1101_0011_1100

1.27

11_1010

1.28

Original Message : Pass 0.12
ASCII encoded message:
1010000_1000001_1010011_1010011_1011100_0110000_0101110_0110001_
0110010|

Encrypted Message after 101 is added:
1010101_1000110_1011000_1011000_1100001_0110101_0110011_0110110_
0110111

1.29

- C. BABBAGE

1.30

- (a) G. BOOLE
- (b) The number of 1s is even in all of the codes. Hence, even parity.

1.31

62 + 32 = 94 printing characters; 34 special characters

1.32 Complement bit 6 (from the right)

1.33

- (a) 8759
- (b) 5426
- (c) F56E
- (d) BEL 、 Y

1.34 ASCII for decimal digits with even parity:

(0): 00110000 (1): 10110001 (2): 10110010 (3): 00110011
(4): 10110100 (5): 00110101 (6): 00110110 (7): 10110111
(8): 10111000 (9): 00111001